

The Compositional Multiplicativity of n -Body Manifolds: Formalizing the κ -Function and Modular Phase-Shift Scaling

Research Memorandum

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Abstract

Recent empirical simulations of 4-body and x -body prime systems have revealed a rigid structural invariant governing composite moduli. This paper formalizes the **κ -Function**, a multiplicative arithmetic function that defines the “curvature” of a manifold. We establish that the previously observed “Manifold Phase Lock” is not a physical failure of the model, but a predictable result of $n \pmod{8}$ divisibility constraints within the M_2 identity. By applying a **Universal Rescaling Protocol**, we demonstrate that the x -body manifold remains analytically continuous across all prime residues.

1 The Fundamental κ -Function

The core discovery defines a new arithmetic function, $\kappa(n)$, which operates as a structural invariant for any composite n . For any prime p , the local curvature is defined as:

$$\kappa(p) = p^2 - p + 1 \quad (1)$$

1.1 The Multiplicative Axiom

For any n composed of distinct primes $\{p_1, p_2, \dots, p_x\}$, the total curvature follows the **Compositional Law**:

$$\kappa(n) = \prod_{i=1}^x (p_i^2 - p_i + 1) \quad (2)$$

This confirms that the curvature is **Interaction-Symmetric**. Whether a 4-body system is analyzed as a $pq|rs$ or $pr|qs$ split, the total curvature $\kappa(pqrs)$ remains invariant.

2 The M_1, M_2 Identities

We define the primary and secondary moments of the manifold as M_1 and M_2 . These moments represent the algebraic signature of the prime set:

- M_1 (**Linear Moment**): $\prod (p_i + 1)$
- M_2 (**Cubic Moment**): $\prod (p_i^3 + 1)$

The relationship between these moments and the curvature is established by the identity:

$$\kappa(n) = \frac{M_2}{M_1} \quad (3)$$

This identity is significant because it allows for the reduction of high-degree cubic interactions into a simpler quadratic curvature field.

3 The Modular Phase-Shift Phenomenon

Empirical data suggests a bifurcation in “Phase Locking” based on the residue of the product n .

3.1 The $n \equiv 1 \pmod{8}$ Resonance

When the product of the x -body system satisfies $n \equiv 1 \pmod{8}$, the derived formula for M_2 yields a clean integer:

$$M_2 = \frac{x - 2nM_1 + M_1}{8} \quad (4)$$

In this state, the manifold is in **Resonance**, and the curvature $\kappa(n)$ is calculated as a clean integer.

3.2 The $n \equiv 5 \pmod{8}$ Half-Integer State

When $n \equiv 5 \pmod{8}$, the phase lock appears to fail. However, our research confirms this is a **Modular Scaling Constraint**. The numerator becomes $4 \pmod{8}$, resulting in a half-integer value (X.5). By applying a factor of 2 to the curvature calculation, the manifold remains solvable.

4 Algebraic Independence & Complexity

While $\kappa(n)$ provides a unique signature, it is functionally dependent on the sum of primes S and the product n . For a 2-body case ($n = pq$):

$$\kappa(pq) = n^2 - n(S - 1) + (S^2 - 2n) - S + 1 \quad (5)$$

This indicates that $\kappa(n)$ provides a **Third Gate**—an additional arithmetic check that must be satisfied by any candidate factor pair.

5 Conclusion

The x -body manifold is governed by a multiplicative curvature law invariant under prime re-grouping. The phase lock failures are resolved through a formal understanding of the $\pmod{8}$ atmosphere. Future research will focus on **Direct Curvature Estimation (DCE)** to calculate $\kappa(n)$ without direct knowledge of prime factors.